TAN SZE AIK

MECHATRONIC ENGINNERING



CONTACT

6010-8337941

tanszeaikaik@gmail.com

www.linkedin.com/in/
szeaik-tan-mechatronics

PROFILE

A Mechatronics Engineering student with interdisciplinary knowledge in electronics, electrical systems, and mechanical engineering, allowing me to approach problem-solving from multiple perspectives. Experienced in automation, circuit analysis and design, and mechanical design through academic studies and hands-on projects. Passionate about developing efficient and sustainable engineering solutions. A determined and persistent individual, I stay committed to my goals and overcoming challenges to continuously improve my skills through practical projects and teamwork.

EDUCATION

MEng (Hons) MECHATRONIC ENGINEERING,
UNIVERSITY OF NOTTINGHAM, MALAYSIA
1st Year – Upper Second Class
2nd Year –

FOUNDATION IN ENGINEERING,
MULTIMEDIA UNIVERSITY (MMU), Malacca
JULY 2023 – AUGUST 2024
CGPA: 3.69

SECONDARY EDUCATION
Pay Fong Middle School, Malacca
January 2017 – December 2022

AWARDS & ACHIEVEMENTS

- **Outstanding Mentor Award** —2026
- **2nd Runner-Up Robotics Competition & Best Robot Arm Design** —2025
- **Course Representative** — 2025
- **S-CPD High Achiever Award** — *2025/26*
- **Dean's List** (Multimedia University) — 2024

SKILLS

Software & Programming :

- SolidWorks
- C Programming
- Adobe Illustrator
- 3D Modeling
- MATLAB
- Phyton

Mechanical & Fabrication :

- Milling machine
- Turning machine
- Lathes machine
- Welding
- Drilling machine
- Marking & Fitting

Language:
English, Chinese, Malay, Cantonese

PROJECT EXPERIENCE

Mapping Microplastic Concentrations in Water Bodies |
Sept 2026 - Present

- ✓ Developing an automated system for spatially mapping microplastic concentrations across water bodies.
- ✓ Exploring the use of sensors and camera images to estimate the amount of microplastics in water.
- ✓ Designing a system to collect and record measurements at different locations.
- ✓ Testing and comparing the results with reference measurements to improve the accuracy of the system.

ICT3 Automatic Sorting Station | *Sept 2025 – May 2026*

- ✓ Designed and assembled an automated sorting system integrating Arduino Mega, Raspberry Pi, sensors, actuators and a 3D-printed robotic arm.
- ✓ Integrated optocouplers and relays to interface 24 V industrial components with 5 V control systems.
- ✓ Developed object detection using a Raspberry Pi camera for automated pick-and-place operations.
- ✓ Designed a robotic arm to grab and place the component using SolidWorks and fabricated using 3D printing.

Centrifugal Clutch | *Sept 2025 – May 2026*

- ✓ Design a brake transmitting 0.3 Nm torque at speeds 2000 rpm.
- ✓ **Project Leader:** Led conceptual design with team and project tutor to refine the design and determine practical manufacturing approaches.
- ✓ Conducted design analysis and prototype fabrication with the team to validate clutch engagement performance.
- ✓ Designed a centrifugal clutch using SolidWorks and manufactured using mechanical machine.

EXTRA-CURRICULAR ACTIVITIES

- SAFAS Certified First Aider
- Sport Coaching
- WEST Program Facilitator

HOBBIES & INTEREST

- Gardening
- Jogging
- Diving
- Fishing

STUDENT LIFE EXPERIENCE

Variable Pitch Propeller | *Sept 2025 – May 2026*

- ✓ Designed a variable pitch marine propeller mechanism focusing on converting linear motion into rotational motion of the propeller blades.
- ✓ Achieved a design capable of providing 10 knots forward speed and 6 knots reverse speed.
- ✓ Developed the mechanical concept and performed design analysis using SolidWorks.

Pit Stop Pneumatic System Project | *Sept 2024 – May 2025*

- ✓ Designed a pneumatic pit stop system in SolidWorks.
- ✓ Focused on the piston and pneumatic actuator design to enable efficient mechanical movement.
- ✓ Manufactured and assembled components using mechanical workshop machines.

SEMICONDUCTOR INTERNSHIP | Infineon Technologies | Malacca | Jun 2026 – Sept 2026 (~16 weeks)

Performed HTOL reliability testing, setting up chambers and monitoring stress conditions for semiconductor devices.

Assisted in equipment management and documentation of testing procedures to enhance lab efficiency.

Troubleshot electrical faults in testing setups & hardware rework to support specific test configurations.

NOTTINGHAM ROBOTIC COMPETITION | *Year 2*

Team Leader: Led design and construction of an autonomous line-following car with robot arm and managed technical issues and optimized performance.

Conducted weight distribution analysis and troubleshooting to improve performance.

Performed strength analysis on the robot arm to ensure durability and reliability.

PEER TO PEER MENTORING PROGRAM | *Year 2*

Served as a mentor, providing academic support to students in 3 modules: Engineering Mathematics, Applied Mechatronic Construct Project, and Analogue, Digital Electronics & Computing.

Applied simulation tools to support student learning and clarify software control of physical systems during the program.

MECHATRONICS REPRESENTATIVE | *Year 1*

Represented students in academic and administrative matters, acting as a communication bridge between students, lecturers, and the department.

Provided timely updates on project deadlines, replacement classes, and class arrangements. Coordinated extra and replacement class rescheduling, including dates, venues, and updates.

Regularly meeting with department and collecting feedback to address learning issues and improve teaching effectiveness.

SAFAS CERTIFIED FIRST AIDER | *Present*

Volunteered as medical support during international events involving participants from more than 5 countries and over 100 attendees.

Collaborated with physiotherapists in a medical support team, delivering first aid assistance and maintaining participant safety throughout the event.